Qizhen Lan

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Houston, TX, United States

Research Interests

- My research focuses on **efficient and trustworthy AI** for computer vision and medical imaging, including knowledge distillation, pruning and acceleration, 3D medical foundation models, and multimodal/agentic systems for clinical and real-world deployment.

Education

- **University of Alabama at Birmingham** — *Jan 2024 – May 2025*
  Ph.D. in Computer Science — Birmingham, United States
- **Bowling Green State University** — *Aug 2019 – Dec 2023*
  Ph.D. student in Data Science — Bowling Green, United States
- **Bowling Green State University** — *Aug 2018 – Jul 2019*
  Master of Science in Analytics — Bowling Green, United States
- **Bowling Green State University** — *Aug 2014 – Jun 2018*
  Bachelor of Science in Business Analytics and Intelligence — Bowling Green, United States

Professional Experience

- **University of Texas Health Science Center at Houston [ ]** — *May 2025 – present*
  Postdoctoral Research Fellow (mentor: Dr. Xiaoqian Jiang) — Houston, TX, United States
  - Pretrained **3D brain foundation models** on large-scale MRI cohorts for Alzheimer's-related downstream tasks.
  - Built a **fast whole-brain segmentation pipeline** with approximately ∼**1s inference**, reducing processing time from FreeSurfer-scale hours to deployment-level latency.
  - Designed region- and context-aware distillation methods for compact 3D segmentation models, achieving **sub-10M-parameter** students while preserving downstream performance.
  - **Outcomes:** One **npj Health Systems** paper accepted and four **MICCAI 2026** papers accepted, including two early accepts (top 9% among 4,601 submissions).

- **University of Alabama at Birmingham (UAB) [ ]** — *Jan 2024 – May 2025*
  Research Assistant (Advisor: Dr. Qing Tian) — Birmingham, AL, United States
  - Led optimization and deployment of object detection models on NVIDIA embedded GPUs (e.g., Jetson Xavier and Orin) for real-time autonomous-driving perception.
  - Developed adaptive **knowledge distillation** methods for dense prediction and DETR-style detectors, reducing model parameters by up to **70%** while maintaining detection performance.
  - **Outcomes**: First-author papers in **ICCV 2025**, **WACV 2026**, and **WACV 2024**, with additional publications in WACV, ICASSP, IV, and T-IV.

- **Bowling Green State University (BGSU) [ ]** — *Aug 2021 – Jan 2024*
  Research Assistant (Advisor: Dr. Qing Tian) — Bowling Green, OH, United States
  - Developed **efficient AI** algorithms for object detection models under autonomous-driving deployment constraints.
  - Built multi-factor and selective dark-knowledge distillation frameworks that reduce model size and computation while preserving detection accuracy across modern detectors.
  - **Outcomes**: IEEE T-IV journal paper and ICPR/WACV publications in efficient object detection and autonomous-driving perception.

SELECTED PUBLICATIONS

J=JOURNAL, C=CONFERENCE, M=MANUSCRIPT

[M.1] **Qizhen Lan**, Yu-Chun Hsu, Nida Saddaf Khan, Xiaoqian Jiang. *Toward Clinically Deployable 3D Auto-Contouring: A Region- and Context-Aware Knowledge Distillation Framework*. **npj Digital Medicine, 2026** (In preparation).

[J.1] **Qizhen Lan**, Aaron Choi, Jun Ma, Bo Wang, Zhongming Zhao, Xiaoqian Jiang, Yu-Chun Hsu. *From Performance to Practice: Knowledge-Distilled Segmentator for On-Premises Clinical Workflows*. **npj Health Systems, 2026**.

[J.2] Jung Im Choi*, **Qizhen Lan***, Qing Tian. *Boosting Deep Detector Efficiency and Robustness through Detection Discriminant Reorganization and Compression*. **Neural Networks, 2026** (* Co-first author; Accepted).

[J.3] **Qizhen Lan**, Qing Tian. *Instance, Scale, and Teacher Adaptive Knowledge Distillation for Visual Detection in Autonomous Driving*. **IEEE Transactions on Intelligent Vehicles (T-IV), 2022**.

[C.1] Zhisheng Chen, Yingwei Zhang, **Qizhen Lan**, Tianyu Liu, Huacan Wang, Yi Ding, Ziyu Jia, Ronghao Chen, Kun Wang, and Xinliang Zhou. *Uni-NTFM: A Unified Foundation Model for EEG Signal Representation Learning*. **ICLR, 2026.**

[C.2] Jinwei Su*, **Qizhen Lan***, Yinghui Xia, Lifan Sun, Weiyou Tian, Tianyu Shi, Xinyuan Song, and Lewei He. *Difficulty-Aware Agentic Orchestration for Query-Specific Multi-Agent Workflows*. **WWW, 2026** (* Co-first author).

[C.3] Tingyu Wu, Zhisheng Chen, Ziyan Weng, Shuhe Wang, Chenglong Li, Shuo Zhang, Sen Hu, Silin Wu, **Qizhen Lan†**, Huacan Wang, and Ronghao Chen. *KnowMe-Bench: Benchmarking Person Understanding for Lifelong Digital Companions*. **ACL Main Conference, 2026** († Corresponding author; Oral Presentation).

[C.4] Xi Xiao, Zhuxuanzi Wang, Chen Liu, **Qizhen Lan**, Lin Zhao, Jianyang Gu, Muchao Ye, Tianyang Wang, and Hao Xu. *CVD-DPO: Anchoring Grounded Reasoning via Counterfactual Visual Decoupling*. **ECCV, 2026.**

[C.5] Mengchen Fan, Baocheng Geng, Xi Xiao, Tianyang Wang, Siyuan Mei, Pulin Che, Xiaoqian Jiang†, and **Qizhen Lan†**. *Detail Consistent Stage-Wise Distillation for Efficient 3D MRI Segmentation*. **MICCAI, 2026** († Corresponding authors; Early Accept).

[C.6] Siyuan Mei, Yanteng Zhang, Yan Xia, **Qizhen Lan**, Yipeng Sun, Siming Bayer, Zirong Li, Chengze Ye, Daiqi Liu, Xiaoqian Jiang, Fuxin Fan, Yixing Huang, and Andreas Maier. *Time Matters: Rethinking Diffusion and Flow Models in One-Step Medical Image Translation*. **MICCAI, 2026** (Early Accept).

[C.7] **Qizhen Lan**, Qing Tian. *CLoCKDistill: Consistent Location-and-Context-aware Knowledge Distillation for DETRs*. **WACV, 2026.**

[C.8] **Qizhen Lan**, Jong-Im Choi, Qing Tian. *Visual Detector Compression via Location-Aware Discriminant Analysis*. **WACV, 2026.**

[C.9] Chen Liu*, **Qizhen Lan***, Zhicheng Ding, Xinyu Chu, Qing Tian. *Learnable Instance Attention Filtering for Adaptive Detector Distillation*. **ICASSP, 2026** (* Co-first author).

[C.10] **Qizhen Lan**, Qing Tian. *ACAM-KD: Adaptive and Cooperative Attention Masking for Knowledge Distillation*. **ICCV, 2025.**

[C.11] Lijing Zhu, **Qizhen Lan**, Qing Tian, Wenbo Sun, Li Yang, Lu Xia, Yixin Xie, Xi Xiao, Tiehang Duan, Cui Tao, Shuteng Niu. *ETT-CKGE: Efficient Task-driven Tokens for Continual Knowledge Graph Embedding*. **ECML-PKDD, 2025.**

[C.12] Yanteng Zhang, Songheng Li, Zeyu Shen, **Qizhen Lan**, Lipei Zhang, Yang Liu, Vince Calhoun. *Lost in Distortion: Uncovering the Domain Gap Between Computer Vision and Brain Imaging–A Study on Pretraining for Age Prediction*. **ISBI, 2025.**

[C.13] Jong-Im Choi*, **Qizhen Lan***, Qing Tian. *Improving Deep Detector Robustness via Detection-Related Discriminant Maximization and Reorganization*. **WACV, 2025** (Oral; * Co-first author).

[C.14] Zhicheng Ding, **Qizhen Lan**, Qing Tian. *Target-Driven and Student-Centered Knowledge Distillation for Traffic Object Tracking*. **IV, 2025.**

[C.15] **Qizhen Lan**, Qing Tian. *Gradient-Guided Knowledge Distillation for Object Detectors*. **WACV, 2024.**

[C.16] Wentao Wang, Xi Xiao, Mingjie Liu, **Qizhen Lan**, Xuanyao Huang, Qing Tian, Swalpa Kumar Roy, Tianyang Wang. *Multi-dimension Transformer with Attention-based Filtering for Medical Image Segmentation*. **ICTAI, 2024.**

HONORS AND AWARDS

- **NSF ACCESS Advanced Computing Allocation (≈$10,000 GPU resources)** *2026*
- **Graduate College Professional Development Funds** *2023*
- **Analytics Instructional Fee Award** *2018*

ACADEMIC SERVICES

- **Conference Reviewer**
  - Annual Conference on Neural Information Processing Systems (NeurIPS), 2026
  - ACM International Conference on Multimedia (ACM MM), 2026
  - International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2026
  - IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026
  - Association for the Advancement of Artificial Intelligence (AAAI), 2025
  - International Symposium on Biomedical Imaging (ISBI), 2025
  - IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2024, 2025
- **Journal Reviewer**
  - IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)
  - Nature - npj Digital Medicine
  - IEEE Transactions on Multimedia (TMM)
  - IEEE Transactions on Neural Networks and Learning Systems (TNNLS)